Smart Home Energy Monitoring App

Complete UI + Flow Specification

Mobile user-side product document with parameter-first, phase-wise real-time and historical monitoring

Core locked rule

Every supported parameter must show Phase 1, Phase 2, Phase 3, and Three Phase / Total together. Every supported parameter must include real-time values, historical data, time filters, graphs, thresholds, and alert history where applicable.

Document Area	Covered
Product scope	Multiple homes, multiple smart energy meters per home, outdoor temperature by home location.
Metrics	Energy Consumption, Voltage, Current, Power, Cost, Power Factor, Frequency, Peak / Max Demand, Active Power, Reactive Power, Cost Per Unit.
Phase structure	Phase 1, Phase 2, Phase 3, and Three Phase / Total shown inside every parameter screen.
Data modes	Real-time and historical data for all parameters.
Cost engine	DISCO tariff categories, slabs, ToU, fixed charges, FPA, QTA, surcharges, taxes, editable versions.
Screens	36 mobile screens including changed cost/tariff screens and parameter detail templates.
Flows	Onboarding, home, meter, parameter, graphs, history, thresholds, alerts, cost, tariff, reports.

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1. Product Scope and Core Rules

Brutally important correction

The UI must not be phase-first. The UI must be parameter-first. Users open Voltage, Current, Power, Cost, etc. and then see Phase 1, Phase 2, Phase 3, and Three Phase / Total inside that parameter with graphs and time filters.

- The app is for homes, not commercial dashboards by default.
- A user can have multiple homes.
- Each home can have one or multiple smart energy meters.
- Each home has its own address, outdoor temperature, tariff setup, thresholds, alerts, and historical data.
- Each meter can be Single Phase or Three Phase.
- For a Three Phase meter, every supported parameter must show Phase 1, Phase 2, Phase 3, and Three Phase / Total.
- Every parameter supports real-time values and historical trends.
- The smart meter measures electrical values. The app calculates cost, projections, alerts, and efficiency insights.
- Do not claim device-level/appliance-level usage unless smart plugs, circuit sensors, or sub-meters are added later.

Entity	Meaning	Important UI Rule
User	Account holder using the mobile app.	Can switch between multiple homes.
Home	A physical location with address and outdoor temperature.	Show city/temperature on dashboard; full address in Home Details.
Meter	Smart energy meter connected to a home.	Can be Single Phase or Three Phase.
Parameter	Measured/calculated value such as Voltage, Current, Power, Cost.	Parameter screen must include all phases + total.
Phase	Phase 1, Phase 2, Phase 3 in a three-phase meter.	Not the main navigation object; phase is inside parameter detail.
Three Phase / Total	Total/sum/average value depending on parameter.	Use correct aggregation per parameter.

2. Information Architecture

```
User Account
└── Homes
    ├── Home Details
    │   ├── Address / City / Country
    │   ├── Latitude / Longitude / Timezone
    │   ├── Outdoor Temperature
    │   ├── DISCO / Tariff Setup
    │   └── Billing Cycle / Budget
    └── Smart Energy Meters
        ├── Single Phase Meter
        │   └── Single Phase / Total values for every parameter
        └── Three Phase Meter
            ├── Phase 1 values for every parameter
            ├── Phase 2 values for every parameter
            ├── Phase 3 values for every parameter
            └── Three Phase / Total values for every parameter
```

Bottom Nav Tab	Purpose	Contains
Home	Fast overview of selected home.	Outdoor temperature, main KPIs, live graph preview, meters, alerts, cost snapshot.
Meters	Meter list and meter details.	Single/three-phase meters, parameter cards, status, latest values.
Usage	Parameter analytics and historical graphs.	Scope, meter, parameter, phase/total, time filters, comparisons.
Alerts	All active and historical warnings.	Threshold alerts, system alerts, event graphs, actions.
More	Settings and deeper tools.	Homes, tariff settings, thresholds, reports, notifications, profile.

3. Parameter-First Monitoring Model

Universal parameter detail rule

Each parameter card opens a parameter detail screen. Each parameter detail screen shows Phase 1, Phase 2, Phase 3, and Three Phase / Total together. Each parameter detail screen includes real-time values, historical graphs, time filters, thresholds, and alerts.

Parameter	Unit	Main/Advanced	Three Phase / Total Aggregation	Primary Graph
Energy Consumption	kWh / units	Main	Sum of Phase 1 + Phase 2 + Phase 3	Stacked bar / line trend
Voltage	V	Main	Average voltage, plus min/max	Multi-line threshold band chart
Current	A	Main	Average current + highest phase current; avoid fake total unless hardware defines it	Multi-line chart with current limit
Power	kW	Main	Sum of phase power	Live multi-line chart
Cost	PKR	Main	Estimated from total consumption; phase costs are estimates	Stacked area / cost trend
Power Factor	ratio	Advanced	Average PF, worst phase	Multi-line chart with quality bands
Frequency	Hz	Advanced	Average frequency	Multi-line threshold band chart
Peak / Max Demand	kW	Advanced	Peak total + peak per phase	Peak bar/line chart
Active Power	kW	Advanced	Sum of phase active power	Multi-line chart
Reactive Power	kVAR	Advanced	Sum of phase reactive power	Multi-line chart
Cost Per Unit	PKR/unit	Advanced / Cost	Current slab rate + average unit cost; phase values are estimated averages	Step/slab chart

3.1 Universal Parameter Screen Template

Section	Required Content
Header	Parameter name, home path, meter name, meter type, status, last sync.
Time Filters	Live, Today, 7 Days, 30 Days, Billing Cycle, Custom.
Current Values	Phase 1 value, Phase 2 value, Phase 3 value, Three Phase / Total value.
Graph	Parameter-specific graph with Phase 1, Phase 2, Phase 3, and Total lines/bars.
Summary	Minimum, maximum, average, total, peak, highest phase depending on parameter.
Thresholds	Active threshold state and threshold line/band on graph when relevant.
Alerts	Recent alert events with marker on graph when possible.
Actions	Set Threshold, View Alerts, Compare Phases, Export/Share if needed.

3.2 Parameter Detail Screens

Parameter Screen	Purpose	Values	Graph
Energy Consumption	Show per-phase and total kWh/units for selected time range.	P1 units, P2 units, P3 units, total units. Include highest phase share.	Stacked bar for phases, total line overlay.
Voltage	Show per-phase voltage and average voltage.	P1 V, P2 V, P3 V, average V, low/high/average values.	Multi-line threshold band chart with low/safe/high zones.
Current	Show per-phase current and average/highest current.	P1 A, P2 A, P3 A, average A, highest phase.	Multi-line chart with overload threshold line.
Power	Show live load per phase and total power.	P1 kW, P2 kW, P3 kW, total kW.	Live multi-line chart with high power threshold.
Cost	Show estimated phase cost and total meter/home cost.	P1 estimated cost, P2 estimated cost, P3 estimated cost, total estimated cost.	Stacked cost area chart with total cost line.
Power Factor	Show PF per phase and average PF.	P1 PF, P2 PF, P3 PF, average PF, worst phase.	Multi-line quality band chart.
Frequency	Show frequency per phase and average frequency.	P1 Hz, P2 Hz, P3 Hz, average Hz.	Multi-line threshold band chart around 50 Hz.
Peak / Max Demand	Show peak demand per phase and total peak.	P1 peak kW, P2 peak kW, P3 peak kW, total peak kW, peak time.	Bar/line chart over selected period.
Active Power	Show real usable power per phase and total.	P1 kW, P2 kW, P3 kW, total kW.	Multi-line chart.
Reactive Power	Show reactive power per phase and total.	P1 kVAR, P2 kVAR, P3 kVAR, total kVAR.	Multi-line chart.
Cost Per Unit	Show tariff-based cost per unit, current slab, and average unit cost.	Current slab rate, average cost/unit, estimated phase average, total meter average.	Step/slab chart + average cost trend.

4. Graph and Time Filter System

Time Filter	Meaning	Recommended Data Granularity
Live	Last 5/15/30 minutes depending on screen.	Raw or near-real-time samples.
Today	Current day from 00:00 to now.	1-min or 5-min summaries.
7 Days	Last seven days.	Hourly summaries.
30 Days	Last thirty days.	Daily summaries + optional hourly drilldown.
Billing Cycle	Current billing cycle based on home tariff settings.	Daily summaries + tariff calculation.
Custom	User-selected date/time range.	System chooses suitable aggregation.

Graph Type	Used For	Required Lines/Bars
Live multi-line chart	Voltage, Current, Power, Active Power, Reactive Power, Frequency, PF.	Phase 1, Phase 2, Phase 3, Three Phase / Total or Average.
Stacked bar chart	Energy consumption by phase over time.	Phase 1, Phase 2, Phase 3 plus total value label.
Stacked area chart	Cost accumulation and phase cost estimate.	Phase 1 estimated cost, Phase 2 estimated cost, Phase 3 estimated cost, total line.
Threshold band chart	Voltage, frequency, power factor, current safety ranges.	Actual values + low/safe/high zones + alert markers.
Slab step chart	Cost per unit and tariff slab movement.	Slab rate steps, average cost/unit line, current slab marker.
Donut / share chart	Phase load share snapshot only.	P1/P2/P3 shares. Use as support, not replacement for parameter graph.

5. Navigation and Global UI Components

Component	UI Behavior
Home selector	User can switch Main Home, Parents Home, Farmhouse, etc. Dashboard recalculates totals.
Meter selector	Filters data by all meters or a specific meter.
Parameter selector	Used in meter/usage screens to switch Energy, Voltage, Current, Power, Cost, etc.
Phase chips	Phase 1, Phase 2, Phase 3, Three Phase / Total. Use for highlighting/toggling lines, not separate-only navigation.
Time filters	Live, Today, 7 Days, 30 Days, Billing Cycle, Custom. Present on all parameter/analytics screens.
Status chips	Online, Delayed, Offline, Warning, Critical, Standard Rates, Manual Override.
Graph legend	Always show P1/P2/P3/Total labels clearly.
Threshold indicators	Show threshold line or band where relevant.

6. Screen Catalogue: 36 Screens

#	Screen	Graphs	Main Purpose
1	Splash	No	Entry/loading screen.
2	Login	No	User authentication.
3	Signup	No	Create account.
4	OTP Verification	No	Verify account.
5	First-Time Setup	No	Prompt to add first home.
6	Add Home	No	Enter home details and address.
7	Confirm Home Location	No	Map pin and outdoor temperature source.
8	Add Meter	No	Scan/enter meter and select meter type.
9	Meter Setup - Single Phase	No	Confirm single-phase parameters.
10	Meter Setup - Three Phase	Yes small	Confirm P1/P2/P3/Total structure.
11	Tariff Category Selection	No	Select DISCO/category/tariff type.
12	Manual Tariff Rates	Yes	Editable slabs/charges preview.
13	Basic Threshold Setup	Yes small	Initial thresholds.
14	Home Dashboard	Yes	Home summary + graphs.
15	All Homes	Yes	Multi-home comparison.
16	Meters List	Yes mini	Meter status and mini trends.
17	Meter Detail - Three Phase	Yes	Parameter cards + selected parameter graph.
18	Meter Detail - Single Phase	Yes	Single-phase parameter graph.
19	Parameter Detail	Yes	Reusable screen for every parameter with P1/P2/P3/Total.
20	Parameter Comparison	Yes	Compare phases for selected parameter.
21	Usage	Yes	Analytics filters and graphs.
22	Historical Parameter Detail	Yes	Deep historical graph + events.
23	Cost & Bill Estimate	Yes	Simple bill projection.
24	Cost Efficiency Calculator	Yes	Budget and daily target.
25	Threshold List	Yes	Threshold summary and health.
26	Create Threshold	Yes	Rule builder + preview graph.

27	Alerts	Yes	Alert list + alert trend.
28	Alert Detail	Yes	Event graph and actions.
29	Reports	Yes	Report generation and preview.
30	More	No	Settings menu.
31	Home Details	Yes	Home profile and usage snapshot.
32	Notification Settings	Yes	Notification settings and activity.
33	Detailed Bill Breakdown	Yes	Full charge/tax breakdown.
34	Tariff Settings	Yes	Standard tariff details and chart.
35	ToU Tariff Settings	Yes	Peak/off-peak rates and schedule.
36	Tariff Version History	Yes	Version timeline and restore.

7. Detailed Screen Specifications

7.1 Onboarding and Setup Screens

Screen	Key UI Details	Validation / States
Splash	Logo, app name, short value proposition.	Auto routes to login or dashboard if already authenticated.
Login	Email/phone, password, login, forgot password, create account.	Invalid credentials, network error, loading state.
Signup	Full name, phone, email, password, confirm password.	Password rules, duplicate email/phone, OTP pending.
OTP Verification	Six-digit OTP fields, resend timer, verify button.	Expired OTP, wrong code, resend success.
First-Time Setup	Message: add home and smart meter to start monitoring.	Skip allowed but dashboard shows empty state.
Add Home	Home name, type, address, city, province, country, map pin.	Address required; location pin recommended for outdoor temperature.
Confirm Home Location	Map preview, selected address, temperature source note.	User can confirm or edit address.
Add Meter	Meter name, QR/manual ID, type, role, assigned home/area.	Invalid meter ID, already linked meter, offline pairing.
Meter Setup - Single Phase	Shows tracked parameters and confirms real-time + historical support.	Confirm setup before dashboard.
Meter Setup - Three Phase	Shows P1/P2/P3/Total and confirms all parameters use this structure.	No fake device-level claims.
Basic Threshold Setup	Daily energy, monthly cost, voltage, power, offline thresholds.	User can skip; default system alerts still active.

7.2 Core Monitoring Screens

14. Home Dashboard

Purpose: Give the selected home status in under 10 seconds.

UI Section	Details
Header	Greeting, selected home dropdown, city/country, outdoor temperature, online/meter/warning chips.
Today Overview	Energy Consumption, Voltage, Current, Power, Estimated Cost.
Cost Snapshot	Current slab, average cost/unit, projected bill.
Meters	Meter cards with type, live power, today units, warning state.
Recent Alerts	Latest critical/warning/info alerts with short path.

Graph	Type	Data / Lines / Bars	Filters
Live Power Preview	Multi-line	P1/P2/P3/Total power	Last 30 min
Phase Load Share	Donut or horizontal bars	P1/P2/P3 share for selected metric, usually Power	Live / Today
Cost Projection	Area/line	Actual cost, projected cost, budget line	Billing Cycle

Rules / States

Dashboard is summary-only. Full parameter graphs open in Parameter Detail.

Show full address only in Home Details, not dashboard.

15. All Homes

Purpose: Let users compare multiple homes quickly.

UI Section	Details
Summary	Total power, today usage, estimated cost, active alerts across all homes.
Home Cards	Home name, city, outdoor temp, meters count, today units, status.
Actions	Add New Home, switch selected home.

Graph	Type	Data / Lines / Bars	Filters
Homes Usage	Bar chart	Usage/cost/power by home	Today / 7 Days / Billing Cycle
Cost Comparison	Line chart	Cost trend per home	7 Days / 30 Days

16. Meters List

Purpose: Show all meters inside selected home.

UI Section	Details
Filters	All, Online, Warning, Offline.
Meter Cards	Name, type, role, assigned area, power, today usage, voltage, current, status.
Actions	View Details, Add Meter.

Graph	Type	Data / Lines / Bars	Filters
Meters Live Load	Stacked/vertical bars	Current power by meter	Live
Mini Power Trend	Small sparkline	Last 1 hour power per meter	1 hour

17. Meter Detail - Three Phase

Purpose: Parameter-first meter monitoring for three-phase meters.

UI Section	Details
Header	Meter name, home, three-phase type, status, last sync.
Time Filter	Live, Today, 7 Days, 30 Days, Billing Cycle, Custom.
Main Parameter Cards	Energy, Voltage, Current, Power, Cost. Each card shows P1/P2/P3/Total summary.
Advanced Parameter Cards	Power Factor, Frequency, Peak Demand, Active Power, Reactive Power, Cost Per Unit.
Selected Parameter Preview	One graph preview for currently selected parameter.
Actions	View Full Graph, Set Threshold, View Alerts.

Graph	Type	Data / Lines / Bars	Filters
Selected Parameter Graph	Depends on parameter	Phase 1, Phase 2, Phase 3, Total	All time filters
Parameter Overview	Compact cards + mini graph	Latest values per phase	Selected time

Rules / States

Do not make Phase 1, Phase 2, Phase 3 the main first-level screen.

Tapping a parameter opens Parameter Detail.

18. Meter Detail - Single Phase

Purpose: Meter monitoring for single-phase meters.

UI Section	Details
Header	Meter name, home, single-phase type, status, last sync.
Time Filter	Live, Today, 7 Days, 30 Days, Billing Cycle, Custom.
Main Parameters	Energy, Voltage, Current, Power, Cost.
Advanced Parameters	Power Factor, Frequency, Peak Demand, Active Power, Reactive Power, Cost Per Unit.
Actions	View Full Graph, Set Threshold, View Alerts.

Graph	Type	Data / Lines / Bars	Filters
Selected Parameter Graph	Line/bar/area	Single Phase / Total value only	All time filters
Historical Summary	Bar chart	Daily consumption/cost etc.	7 Days / 30 Days

19. Parameter Detail

Purpose: Reusable full screen for every parameter.

UI Section	Details
Header	Parameter name, home -> meter path, meter type, status.
Time Filter	Live, Today, 7 Days, 30 Days, Billing Cycle, Custom.
Values	P1, P2, P3, Three Phase / Total or Average depending on parameter.
Summary	Min, max, average, total, peak, highest phase as applicable.
Alerts	Recent events for selected parameter.
Actions	Set Threshold, View Alerts, Compare Phases.

Graph	Type	Data / Lines / Bars	Filters
Parameter Trend	Parameter-specific	Phase 1 / Phase 2 / Phase 3 / Total	All time filters
Threshold Overlay	Band/line overlay	Safe/warning/critical ranges where relevant	Selected time

Rules / States

This is the most important screen pattern in the app.

All parameter screens must use the same phase-and-time-filter structure.

20. Parameter Comparison

Purpose: Compare all phases for one selected parameter.

UI Section	Details
Selector	Parameter selector: Energy, Voltage, Current, Power, Cost, etc.
Values	P1, P2, P3, Total/Average values.
Insight	Highest/worst phase, imbalance or abnormal condition.
Actions	Create threshold, view full parameter graph.

Graph	Type	Data / Lines / Bars	Filters
Grouped Comparison	Grouped bar	P1/P2/P3/Total values	Live / Today / 7 Days / Billing Cycle
Phase Share	Donut, only for sum-based metrics	P1/P2/P3 share	Selected range

21. Usage

Purpose: Explore historical analytics by home, meter, parameter, phase/total, and time.

UI Section	Details
Scope Filters	All Homes, selected home, selected meter.
Parameter Selector	All supported parameters.
Phase Toggle	Show/hide P1, P2, P3, Total lines.
Time Filter	Today, 7 Days, 30 Days, Billing Cycle, Custom.
Comparison	Today vs yesterday, this month vs last month, phase comparison.

Graph	Type	Data / Lines / Bars	Filters
Main Usage Chart	Bar/line/area depending on parameter	P1/P2/P3/Total	All time filters
Phase Trend	Stacked area for sum metrics	Phase contribution over time	Today / 7 Days / Billing Cycle

22. Historical Parameter Detail

Purpose: Deep history screen for any parameter.

UI Section	Details
Header	Parameter history, selected home/meter, selected phase visibility.
Stats	Current, lowest, highest, average, event count.
Events	Alerts/normalization events listed under graph.
Actions	Set threshold, export/share if needed.

Graph	Type	Data / Lines / Bars	Filters
Historical Trend	Threshold band line chart	P1/P2/P3/Total + alert markers	Today / 7 Days / 30 Days / Custom

7.3 Cost and Tariff Screens

11. Tariff Category Selection

Purpose: Choose provider, residential category, tariff type, and billing cycle.

UI Section	Details
Home	Selected home.
Provider / DISCO	IESCO, LESCO, FESCO, GEPCO, MEPCO, PESCO, HESCO, SEPCO, QESCO, TESCO, K-Electric, Other.
Consumer Category	Default home flow shows A-1 Residential only.
Tariff Type	A-1a Lifeline, A-1a Protected, A-1a Non-Protected, A-1b ToU Residential.
Calculation Method	Auto-detected as Slab or ToU.
Rates Source	Use Standard Rates or Manual Override.

Rules / States

Commercial/industrial categories should be hidden unless advanced mode is enabled.

If ToU is selected, route to ToU Tariff Settings.

34. Tariff Settings

Purpose: Show standard or active tariff details for selected home.

UI Section	Details
Header	Home, provider, category, tariff type, calculation method.
Rate Source	Standard Rates or Manual Override, effective date, last updated.
Slab Rates	All residential slabs displayed with rate/unit.
Other Charges	Fixed charges, FPA, QTA, FC surcharge, GST, Income Tax, TV fee, LPS etc.
Actions	Edit Rates, Version History, Save Tariff Version.

Graph	Type	Data / Lines / Bars	Filters
Tariff Slab Chart	Stepped line chart	Unit range vs rate/unit	Tariff version

Rules / States

Fields locked for standard rates until user taps Edit Rates.

Save changes as a new tariff version.

12. Manual Tariff Rates

Purpose: Manual override editor for slabs, taxes, and charges.

UI Section	Details
Slabs	0-100, 101-200, 201-300, 301-400, 401-500, 501-600, 601-700, 701+.
Charges	Fixed, FPA, QTA, FC surcharge, N.J. surcharge.

Taxes	GST, Further Tax, Extra Tax, Income Tax, Electricity Duty, TV Fee, Late Payment Surcharge.
Version	Effective date and optional version note.
Actions	Save Override.

Graph	Type	Data / Lines / Bars	Filters
Live Tariff Preview	Stepped slab chart	Updated rates as user edits	Current edit state

Rules / States

This is an advanced screen. Do not show it by default to normal users.

35. ToU Tariff Settings

Purpose: Configure peak/off-peak residential ToU tariff.

UI Section	Details
Rates	Peak rate, off-peak rate, fixed charge.
Adjustments / Taxes	FPA, QTA, FC surcharge, GST, Income Tax, TV fee.
Peak Hours	Winter 5 PM-9 PM, Summer 6 PM-10 PM or editable by admin/user.
Source	Standard Rates or Manual Override.
Actions	Save.

Graph	Type	Data / Lines / Bars	Filters
Peak vs Off-Peak Rate	2-column comparison	Peak rate and off-peak rate	Tariff version

Rules / States

For ToU cost, backend must store peak_units and off_peak_units separately.

23. Cost & Bill Estimate

Purpose: Simple user-facing estimated bill screen.

UI Section	Details
Header	Home, billing cycle, provider, tariff type.
Estimated Bill	Estimated payable, units used, current slab, current slab rate, average cost/unit.
Bill Summary	Energy charges, fixed charges, taxes/adjustments.
Budget	Budget, current projection, over/under budget status.
Actions	View Detailed Bill, Edit Tariff, Set Cost Alert.

Graph	Type	Data / Lines / Bars	Filters
Cost Projection	Area/line	Actual cost, projected bill, budget line	Billing Cycle
Slab Progress	Progress chart	Completed/current/upcoming slabs	Current cycle

Rules / States

Call it Estimated Bill, not exact bill unless all bill components are fully included.

For ToU, replace slab progress with peak/off-peak cost split.

33. Detailed Bill Breakdown

Purpose: Advanced screen showing full bill engine output.

UI Section	Details
Consumption Summary	Total units, current slab, peak/off-peak units if ToU.
Energy Charges	Per-slab or peak/off-peak line items.
Utility Charges	Fixed charges, FPA, QTA, utility subtotal.

Government Charges	FC surcharge, N.J. surcharge, GST, Further Tax, Extra Tax, Income Tax, Electricity Duty, TV Fee, Taxes on FPA.
Final Bill	Payable within due date, late payment surcharge, payable after due date.
Actions	Export Breakdown, Edit Rates.

Graph	Type	Data / Lines / Bars	Filters
Bill Composition	Stacked bar/donut	Energy, fixed, taxes, adjustments	Billing Cycle
Slab or ToU Breakdown	Bar chart	Per-slab cost or peak/off-peak cost	Current cycle

24. Cost Efficiency Calculator

Purpose: Help user stay within budget using tariff-aware estimates.

UI Section	Details
Inputs	Monthly budget, billing cycle, current units, estimated bill so far.
Result	Budget remaining, allowed daily spend, allowed daily units, current daily average, required reduction.
Insight	Warn if user is moving into higher-cost slab or projected over budget.
Actions	Create Budget Alert, Edit Budget.

Graph	Type	Data / Lines / Bars	Filters
Budget Progress	Progress ring	Used budget %, remaining budget %	Billing Cycle
Daily Usage Target	Line chart	Allowed daily units vs actual daily units	Billing Cycle
Cost Accumulation	Line/area	Actual cost, projected cost, budget line	Billing Cycle

36. Tariff Version History

Purpose: Track tariff changes so historical costs remain explainable.

UI Section	Details
Current Version	Version number, effective date, source, active status.
Previous Versions	Version list with date, source, reason.
Version Detail	Bottom sheet showing changed slabs, charges, taxes.
Actions	View details, restore old version if allowed.

Graph	Type	Data / Lines / Bars	Filters
Version Timeline	Timeline strip	Version 1 -> Version 2 -> Version 3 -> Current	Home / tariff type

Rules / States

Bill estimates must store tariff_version_used so old reports do not change when rates are updated.

7.4 Thresholds, Alerts, Reports, and Settings Screens

25. Threshold List

Purpose: Show all basic and advanced threshold rules.

UI Section	Details
Tabs	Basic and Advanced.
Cards	Daily Energy, Monthly Cost, Low Voltage, High Power, Phase Imbalance, Meter Offline.
Each Card	Scope, parameter, condition, value, status, mini graph.
Action	Create Threshold.

Graph	Type	Data / Lines / Bars	Filters
Threshold Health	Donut	Active, triggered, muted, resolved	Selected home
Mini Graphs	Sparkline/band	Current value vs threshold	Rule-specific

26. Create Threshold

Purpose: Build notification rule for a parameter.

UI Section	Details
Scope	Home, Meter, Phase 1, Phase 2, Phase 3, Three Phase / Total.
Parameter	All supported parameters plus Phase Imbalance and Meter Offline.
Condition	Greater than, less than, between, outside range, percentage difference, no data received.
Rule Details	Value, unit, duration, severity, notification, cooldown.
Action	Save Threshold.

Graph	Type	Data / Lines / Bars	Filters
Threshold Preview	Band/line preview	Safe/warning/critical zones + selected threshold line	Current configuration

Rules / States

For parameter thresholds, user must choose scope and phase/total context clearly.

27. Alerts

Purpose: Show alert feed with trend and quick actions.

UI Section	Details
Filters	Home, All/Critical/Warning/Resolved.
Cards	Alert type, home -> meter -> parameter/phase path, actual value, threshold, time, actions.
Actions	Acknowledge, View, Resolve, Create Rule.

Graph	Type	Data / Lines / Bars	Filters
Alert Trend	Bar chart	Critical, warning, info count by day	Last 7 / 30 days
Mini Alert Graph	Sparkline	Parameter value near alert time	Event window

28. Alert Detail

Purpose: Explain exactly what happened and what user can do.

UI Section	Details
Header	Alert type, severity, home, meter, parameter, phase/total.
Event	Actual value, threshold, started at, duration, status.
History	Timestamped event progression.
Actions	Acknowledge, Resolve, Mute, Edit Threshold.

Graph	Type	Data / Lines / Bars	Filters
Alert Event Chart	Line with threshold	Actual value, threshold line, alert marker	Event timeline

29. Reports

Purpose: Generate and view reports.

UI Section	Details
Filters	Home, meter, parameter, phase/total, report type, date range.
Recent Reports	Monthly Cost, Phase/Parameter Comparison, Voltage Quality, Alert Report.
Actions	Generate Report, Download/Share later.

Graph	Type	Data / Lines / Bars	Filters
Report Preview	Multi-chart preview	Energy, cost, parameter comparison, voltage quality, alert count	Selected report range

30. More

Purpose: App settings hub.

UI Section	Details
Account	Profile, Notification Settings.
Homes & Meters	My Homes, Meters, Add Home, Add Meter.

Energy Settings	Tariff Settings, Thresholds, Cost Calculator, Reports.
Support	Help Center, Contact Support.
App	Privacy, Terms, Logout.

31. Home Details

Purpose: Display and manage selected home profile.

UI Section	Details
Profile	Home type, address, outdoor temperature, meters connected.
Energy Settings	DISCO, tariff, billing cycle, monthly budget.
Actions	Edit Home, Manage Meters, Tariff Settings.

Graph	Type	Data / Lines / Bars	Filters
Home Usage Snapshot	Area chart	Total home usage last 7 days	7 days
Meter Split	Donut	Usage split by meter	Today / Billing Cycle

32. Notification Settings

Purpose: Control notification channels and cooldowns.

UI Section	Details
Channels	Push, in-app, later SMS/email/WhatsApp.
Critical Alerts	Always on.
Quiet Hours	Start/end time.
Cooldown	Avoid repeated spam.
Alert Types	High cost, high energy, voltage, current, power, offline, imbalance, frequency, PF.

Graph	Type	Data / Lines / Bars	Filters
Notification Activity	Bar chart	Critical/warning/info notifications sent	Last 7 days

8. Cost and Tariff Engine

Source logic from uploaded calculator

The uploaded IESCO calculator uses category-based tariff methods: slab, flat, and Time of Use. It also includes editable rates, fixed charges, FPA, QTA, FC surcharge, N.J. surcharge, GST, Further Tax, Extra Tax, Income Tax, Electricity Duty, TV Fee, taxes on FPA, and late payment surcharge. The mobile app should expose simple estimate first, then detailed breakdown on demand.

Calculation Method	Formula	Where Used
Slab	Sum of units_in_slab x slab_rate for each slab incrementally.	A-1 residential lifeline/protected/non-protected.
Flat	Total units x flat rate.	Non-home or advanced future categories.
Time of Use	Off-peak units x off-peak rate + peak units x peak rate.	A-1b ToU and future ToU meters.
Fixed charge	Fixed amount, phase-based amount, or MDI x fixed rate depending on tariff.	Residential simple, industrial/commercial future.
FPA	Previous month units x FPA rate.	Detailed bill estimate.
QTA	Current units x QTA rate.	Detailed bill estimate.
Tax base	Energy + fixed + QTA + FC surcharge, based on calculator logic.	Government charges.
Final payable	Utility subtotal + government subtotal; late payable adds LPS.	Detailed Bill Breakdown.

User-Facing Label	Meaning	UX Rule
Estimated Bill	Calculated bill estimate from live/historical meter data + tariff version.	Use this label unless exact official bill is guaranteed.
Current Slab Rate	Rate for the slab the user is currently in.	Different from average cost/unit.
Average Cost Per Unit	Estimated total cost divided by units.	Useful for efficiency comparisons.
Projected Bill	Forecast to end of billing cycle.	Must update as consumption changes.
Detailed Bill Breakdown	Advanced bill line-items.	Hidden behind View Detailed Bill.
Tariff Version	Snapshot of rates used for calculation.	Required for historical reports.

9. Thresholds, Alerts, and Notifications

Rule Scope	Examples
Home	Monthly cost crosses PKR 25,000; daily usage crosses 20 units.
Meter	Main Meter power crosses 5 kW for 10 minutes.
Parameter	Voltage below 190V; frequency outside range; PF below threshold.
Phase 1 / Phase 2 / Phase 3	Phase 1 current crosses 20A; Phase 2 voltage drops below 190V.
Three Phase / Total	Total power crosses 8 kW; total cost crosses budget.

Alert Type	Trigger Logic	Recommended Severity
High Energy Consumption	Energy exceeds daily/monthly user threshold.	Warning
High Projected Bill	Projected bill exceeds budget.	Warning
Entered Next Slab	Consumption crosses into higher tariff slab.	Info/Warning
Low Voltage	Voltage below configured safe value for duration.	Warning/Critical
High Voltage	Voltage above configured safe value for duration.	Warning/Critical
High Current	Current exceeds configured limit.	Warning/Critical
High Power	Power exceeds configured limit for duration.	Warning
Phase Imbalance	One phase share/difference exceeds configured threshold.	Warning
Meter Offline	No data received for configured duration.	Critical
Frequency Abnormal	Frequency outside safe range.	Warning
Low Power Factor	PF below threshold.	Info/Warning

10. End-to-End Application Flows

Flow	Steps
First-Time Onboarding	Open app -> Signup/Login -> Add Home -> Confirm Location -> Fetch Outdoor Temperature -> Add Meter -> Select Single/Three Phase -> Configure Tariff -> Set Basic Thresholds -> Dashboard.
Add Home	More/Home selector -> Add Home -> Enter details -> Confirm map pin -> Fetch weather -> Set DISCO/tariff -> Add meter now or later.
Add Meter	Meters -> Add Meter -> Scan QR/enter ID -> Select home -> Select meter type -> Select role -> Confirm parameter support -> Set thresholds -> Meter detail.
Three-Phase Parameter Monitoring	Meters -> Main Meter -> Tap parameter card (Voltage/Current/Power/etc.) -> Parameter Detail -> P1/P2/P3/Total values -> Select time filter -> View graph -> Set threshold or view alerts.
Real-Time Data	Meter sends reading -> Cloud validates -> Store latest reading -> Update dashboard/parameter screen -> Check thresholds -> Create alert if matched -> Push/in-app notification.
Historical Data	Usage -> Select home/meter/parameter -> Choose P1/P2/P3/Total visibility -> Select time range -> Load aggregated data -> Show graph + summary + events.
Cost Estimate	Cost screen -> Read billing cycle units -> Apply active tariff version -> Calculate slab/ToU + charges/taxes -> Show estimated bill -> Project month-end bill.
Tariff Setup	Tariff Category -> Select DISCO/category/type -> Load standard rates -> User may edit -> Save tariff version -> Use version in all future estimates.
Threshold Setup	Thresholds -> Create -> Select scope -> Select parameter -> Condition/value/duration -> Severity/channel/cooldown -> Save -> Evaluate against incoming readings.
Alert Handling	Alert triggered -> Alert card appears -> User views detail -> Checks graph/history -> Acknowledge/Resolve/Mute/Edit threshold.
Meter Offline	No data -> delayed state -> no data beyond threshold -> offline alert -> data returns -> reconnected event.
Report Generation	Reports -> Select home/meter/parameter/phase/time -> Generate -> Include real-time summary + historical chart + tariff version + alert summary.

11. Data Model Requirements

Model	Required Fields / Notes
User	id, name, phone, email, auth metadata.
Home	id, user_id, name, type, address, city, province, country, lat, lng, timezone, outdoor_temp, weather_updated_at, DISCO, tariff_plan, billing_cycle_day, budget.
Meter	id, home_id, name, meter_id, type single/three, role, status, last_sync_at.
MeterReading	meter_id, timestamp, all parameters for P1/P2/P3/Total, tariff version active, estimated cost fields.
ParameterSummary	minute/hour/day/month summaries for every parameter and every phase/total.
TariffVersion	provider, category_code, method slab/flat/tou, effective_from/to, source, standard/manual, version note.
TariffRateSlab	version_id, from_units, to_units, rate_per_unit, label.
TariffCharge	version_id, charge_type, calculation_type, rate, fixed_amount, applies_to.

TaxRule	GST, Further Tax, Extra Tax, Income Tax, Electricity Duty, taxes on FPA.
BillEstimate	home_id, meter_id optional, cycle, units, peak/offpeak, current slab, avg cost/unit, projected bill, tariff_version_used.
ThresholdRule	scope, home_id, meter_id optional, parameter, phase_or_total, condition, value, duration, severity, cooldown, active.
Alert	type, home_id, meter_id, parameter, phase_or_total, actual value, threshold, severity, status, created/ack/resolved timestamps.

11.1 Meter Reading Field Pattern for Three Phase

For every supported parameter, store:

- phase_1_<parameter>
- phase_2_<parameter>
- phase_3_<parameter>
- total_or_average_<parameter>

Examples:

- phase_1_voltage, phase_2_voltage, phase_3_voltage, avg_voltage
- phase_1_power_kw, phase_2_power_kw, phase_3_power_kw, total_power_kw
- phase_1_energy_kwh, phase_2_energy_kwh, phase_3_energy_kwh, total_energy_kwh
- phase_1_estimated_cost, phase_2_estimated_cost, phase_3_estimated_cost, total_estimated_cost

12. Figma Design Notes

- Start with 10 critical screens: Home Dashboard, Meters List, Meter Detail - Three Phase, Parameter Detail, Usage, Cost & Bill Estimate, Detailed Bill Breakdown, Tariff Settings, Threshold List, Alerts.
- Create one reusable Parameter Detail component/template and duplicate it for Energy, Voltage, Current, Power, Cost, Power Factor, Frequency, Peak Demand, Active Power, Reactive Power, and Cost Per Unit.
- Use phase legend consistently: Phase 1, Phase 2, Phase 3, Three Phase / Total.
- Use chips to show/hide phase lines on graphs.
- Use a light mobile UI: white cards, grey background, electric blue accents, green normal, amber warning, red critical.
- Do not show the detailed bill calculator on the dashboard. Dashboard should show only cost snapshot and projection.
- Do not use “Digital Twin” wording in user-facing mobile UI. Keep backend/system structure hidden under simple home/meter language.

Color Token	Hex	Use
Primary Blue	#2563EB	Buttons, selected chips, active graph line.
Deep Navy	#0F172A	Primary text.
Light Background	#F8FAFC	App background.
Card White	#FFFFFF	Cards.
Border	#E2E8F0	Card borders/dividers.
Normal Green	#22C55E	Normal states.
Warning Amber	#F59E0B	Warning states.
Critical Red	#EF4444	Critical alerts.